Sequenzanalyse 2

by Jens Stoye

Summer semester 2026

Lecture notes by Liliana Sanfilippo

April 2026

Contents

CHAPTER 1

Sequence Comparison Beyond Edit Distance

The edit distance (or alignment distance) is one of the most fundamental ways of quantifying the dissimilarity of two sequences x and y over a finite alphabet $\sum$. It is based on the cost of an alignment $A = (A_1, A_2, ..., A_n)$, , defined as the sum of the costs of A's columns, i.e.,

$$cost(A) = \sum_{i=1}^{n} cost(A_i),$$

where the cost of alignment column $A_i = \binom{a_i}{b_i}$, $a_i, b_i \in \sum$, in the simplest (unit cost) case is 0 for a match column $(a_i = b_i)$, and 1 for a mismatch column $(a_i \neq b_i)$ or an indel column $(a_i = -orb_i = -)$. Then we have:

Definition 1.1 alignment distance
The alignment distance of two sequences $x, y \in \sum^*$ is defined as $d(x, y) := mincost(A) : A \in \mathcal{A} *$ is an alignment of x and y

The corresponding computational problem is as follows:

Problem 1.1 Alignment Problem
For two given strings $x, y \in \sum^*$ and a given cost function, find the alignment distance of x and y and one or all optimal alignment(s).

From the "Sequence Analysis 1" lecture we know that this problem can be solved in $O(mn)$ time using a simple and elegant dynamic programming algorithm, where $m = |x|$ and $n = |y|$ are the two sequence lengths.

However, there are situations in which edit distance and alignments are not the perfect model to compare sequences. In the following two sections we will study two interesting alternatives.

Hier das über die Block moves einfügen.

1.1 | The q-gram Distance

Edit distances emphasize the correct order of the characters in a string. If, however, a large block of a text is moved somewhere else (e.g. two paragraphs of a text are exchanged), the edit distance will be high, although from a certain standpoint, the texts are still quite similar. A more appropriate notion of distance can be defined in several ways. Here we introduce the q-gram distance dq, which is actually a pseudo-metric: There exist strings $x \neq y$ with $d_q(x, y) = 0$.

Remember that for $q \in N$, a q-gram is a string of length q. Wir reden hier über überlappende q-Gramme!

Definition 1.2 occurrence count
Let $x \in \sum^m$ and choose $q \in [1, m]$. The occurrence count of a q-gram $z \in \sum^q$ in x is the number

$$Occ(x, z) := |i \in [1, m - q + 1] : x[i...i + q - 1] = z|$$

Definition 1.3 q-gram profile
Let $\sigma = |\sum|$. The q-gram profile of x is a σ^q-element vector $p_q(x) := (p_q(x)_z)_{z \in \sum^q}$, indexed by all q-grams, defined by $p_q(x)_z := Occ(x, z)$

Definition 1.4 q-gram distance
The q-gram distance of $x \in \sum^m$ and $y \in \sum^n$ is defined for $1 \leq q \leq min\{m, n\}$ as

$$d_q(x, y) := \sum_{z \in \sum^q} |p_q(x)_z - p_q(y)_z|$$

Beispiel 1.1 Consider $\sum = A, B$, $x = ABAA$, $y = ABAB$, $z = AABA$, and $q = 2$. Using lexicographic order (AA, AB, BA, BB) among the q-grams, we have $p_2(x) = (1, 1, 1, 0)$, $p_2(y) = (0, 2, 1, 0)$ and $p_2(z) = (1, 1, 1, 0)$. Therefore we obtain $d_2(x, y) = 2$ and $d_2(x, z) = 0$ (although $x \neq z$). Hier bsp. tabelle

Theorem 1.1 The q-gram distance dq is a pseudo-metric.

Proof. It fulfills nonnegativity, symmetry and the triangle inequality (exercise).

1.1.1 | Practical Computation

If the q-gram profile of a string $x \in \sum^m$ is (i.e., if it does not contain mostly zeros), it makes sense to implement it as an array $p[0 \cdots \sum^q - 1]$, where $p[i]$ counts the number of occurrences of the i-th q-gram in some arbitrary but fixed (e.g. lexicographic) order.

Definition 1.5 ranking function

Definition 1.6 ranking algorithm

Definition 1.7 unranking function

Definition 1.8 unranking algorithm

Of course, we are interested in an efficient (un-)ranking algorithm for the set of all q-grams over $\sum$. A classical one is to first define a ranking function $r_\sum$ on the characters (alphabet encoding) and then extend it to a ranking function r on the q-grams by interpreting them as base-σ numbers. There are two possibilities to number the positions of a q-gram: From q to 1 falling or from 1 to q rising, as shown in the Example 1.2.

Beispiel 1.2 Assume the alphabet $\sum = A, C, G, T$ and the alphabet encoding $r_\sum(A) = 0$, $r_\sum(C) = 1, r_\sum(G) = 2$, and $r_\sum(T) = 3$. For q = 5, the two different possibilities to rank the q-gram z = ATACG are:

1. Falling ranking
$$r(ATACG) = ...$$
2. Rising ranking
$$r(ATACG) = ...$$

The first numbering of positions in (a) corresponds naturally to the ordering of positional number systems; in the decimal number 23, the first 2 has positional value (weight) $10^1 = 10$ whereas the 3 has the lower weight of $10^0 = 1$. For texts, however, the second numbering in (b) is more natural since we expect lower position numbers on the left

side of higher position numbers.

In general, we see that $z \in \sum^q$ has rank

$$(a) : r(z) = \sum_{i=1}^{q} r_{\sum}(z[i]) \cdot \sigma^{q-i}, (b) : r(z) = \sum_{i=1}^{q} r_{\sum}(z[i]) \cdot \sum^{i-1}.$$

Whichever numbering system we use, we have to do it consistently.

For each of the two ways to rank a q-gram, there is a corresponding unranking method. Both methods work with integer division and remainder. The one for the falling ranking function uses a dynamic divisor and determines the characters of the q-gram based on the integer results, the other one for the rising ranking function uses a fixed divisor and determines the characters based on the remainder

1.1.2 | Choice of q

1.1.3 | Efficiency

1.1.4 | Relation to the Edit Distance

1.2 | The Maximal Matches Distance

Notiz

CHAPTER 2

Practical Aspects of de Bruijn Graphs

A note on terminology: In Section 1.1 we were speaking of q-grams when referring to short (sub)strings of a certain length q. This notation has originally been introduced in the computer science literature and is there in use until today. An alternative, equivalent notion that one finds in most of the biological[i] and bioinformatics literature, is that of a k-mer. Here k refers to the (sub)string length. To be more consistent with the literature, in this chapter we will adopt that notation.

2.1 | Definition and Basic Properties

Remember the following definition, where $s_{k,i} \coloneqq s[i \dots i+k-1]$ denotes the k-mer starting at index i, $1 \leq i \leq |s|-k+1$:

Definition 2.1 **de Bruijn subgraph**

Definition 2.2 **dimension of a de Bruijn subgraph**

Note 2.1 **To 2.1** If the same (k + 1)-mer occurs multiple times in s, say m times, we have m parallel edges occurring in the multigraph DBG(s, k).

Kommentar 2.1 Für jeden DB Graph gibt es immer einen Eulerpfad

2.2 | Words with the same (k + 1)-mer Profile

Motivated by the fact that the q-gram distance does not satisfy the identity property of a metric (see Section 1.1) because there can be distinct sequences with the same q-gram profile, we want to determine whether for a given sequence s and integer k there are other sequences that are distinct but have the same (k + 1)-mer profile as s. And, if there are, how many?

For $k = 0$, the problem is trivial. Indeed, if a sequence t that is distinct from s corresponds to a permutation of the symbols of s, then s and t clearly share the same 1-mer profile. For $k \geq 1$, an elegant answer can be found with the help of the de Bruijn subgraph DBG(s, k).

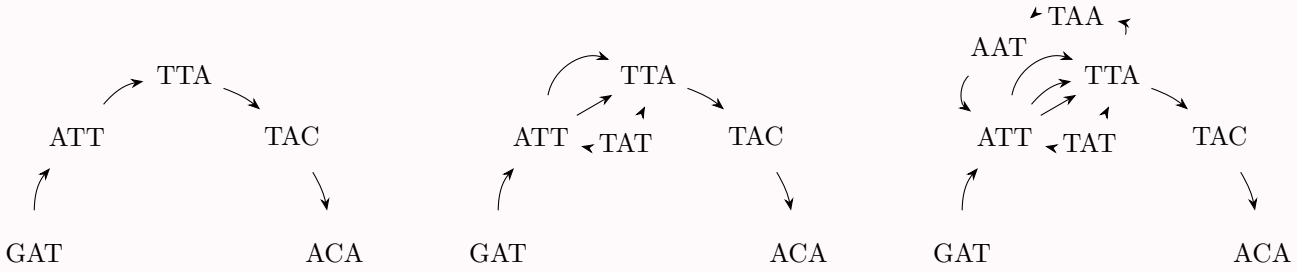
[i] The true origin is probably in biochemistry, where notions like monomer or polymer refer to molecular chains of certain lengths.

Clearly, by construction DBG(s, k) forms an Eulerian path p_s , represented by its sequence of vertices

$$s_{k,1}, s_{k,2}, \ldots, s_{k,|s|-k+1}.$$

Consequently, DBG(s, k) is connected and balanced and might contain even more than one Eulerian path. In fact, there is a one-to-one correspondence between Eulerian paths in DBG(s, k) and sequences sharing the same (k + 1)-mer profile. In other words, a sequence t that is distinct but has the same (k + 1)- mer profile as s exists if and only if DBG(s, k) has another Eulerian path p_t orresponding to the sequence of (k + 1)-mers of t. Since p_t uses the same edges as p_s n a different order, the path p_t an only exist if DBG(s, k) contains a cycle. Note, however, that the presence of a cycle is not a sufficient condition. Indeed, it is possible that DBG(s, k) contains one or more cycles, but still admits only one Eulerian path. Some examples are given in Figures 2.1 and

Example 2.1



(a) $x = GATTACA, k = 3$ - Since this graph contains no cycle, it has only one Eulerian path. There can be no other word sharing the same 4-mer profile.

(b) $x = GATTATTACA, k = 3$ - Here the graph contains two cycles, but still only one Eulerian path. There is no other word sharing the same 4-mer profile.

(c) $x = GATTATTAATTACA, k = 3$ - This graph contains several cycles and two distinct Eulerian paths. There is exactly one other word with the same 4-mer profile, `GATTAATTATTACA`.

Figure 2.1: Examples of de Bruijn subgraphs for distinct sequences and k = 3.

Example 2.2 Additional example from lecture

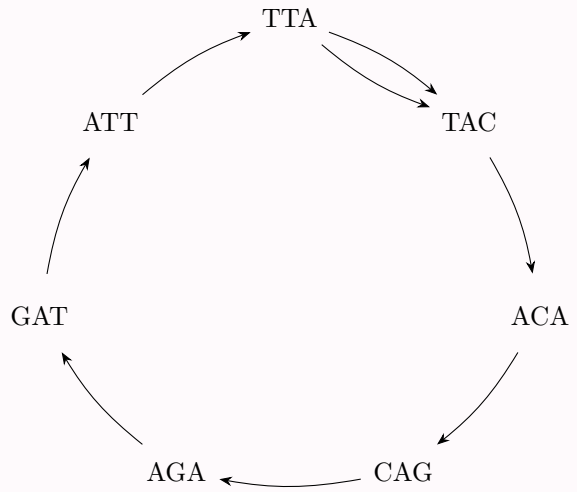


Figure 2.2: It could also be on big circle.

Kommentar 2.2 Reminder that the edges are basically this:

Kommentar 2.3 **To figure 2.1** man könnte bei fig 2.1a denken hey bei a kann es nur einen EP geben, weil es kein Kreis ist, und bei 2.1b muss es dann mehrere geben, weil da eine Schleife drin ist, aber NEIN der Pfad ist die Reihenfolge der Knoten und nicht der Kanten, also gibt es nur einen. Man kann nicht unterscheiden, ob man zuerst die obere oder untere Kante der Schleife zuerst gegangen ist. Gibt also nur eine sequenz mit em k+1 meer profil. Denn die Sequenz wird ja aus den Knoten gemacht, nicht den Kanten …sozusagen.

We summarize the observations above in the following theorem.

Theorem 2.1 Given a sequence s and an integer $k \geq 1$, the number of distinct sequences that have the same (k + 1)-mer profile as s equals the number of distinct Eulerian paths in DBG(s, k). If DBG(s, k) is acyclic, it has a single Eulerian path, and no sequence t exists that is distinct but has the same (k + 1)-mer profile as s.

Kommentar 2.4 Auf deutsch: # worte mit gleichen (k+1)-mer Profil wie s = # Euler-Pfade in DBG(s,k)

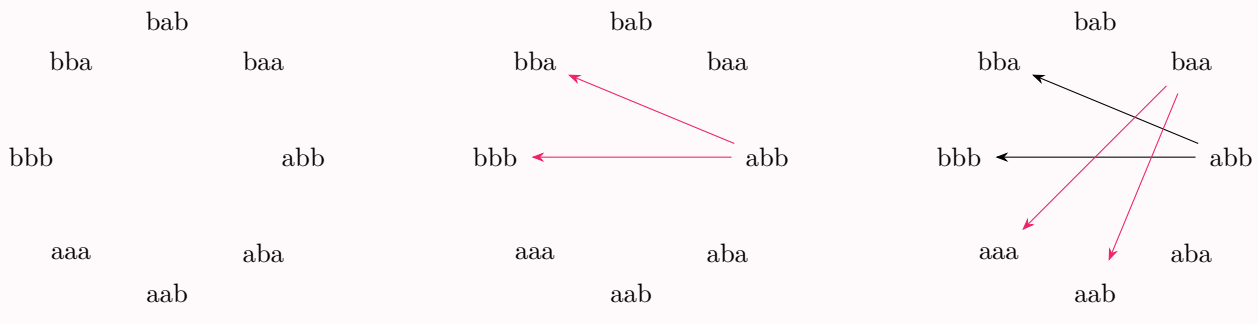
2.3 | Variants of de Bruijn Graphs

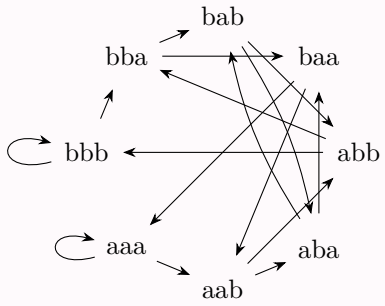
Kommentar 2.5 Der Begriff dbg ist sehr dehnbar, es ist in der Literatur nicht immer das gleiche damit gemeint. Viele sagen dbg, wenn sie eigentlich dbsg meinen.

Although, strictly speaking, we consider most of the time de Bruijn subgraphs (for a given sequence), often they are just called de Bruijn graphs. In fact, there are more variants of de Bruijn graphs, and often they are also not named very carefully. Here we will discuss them and introduce a precise nomenclature.

Definition 2.3 **original de Bruijn graph of dimension k over a finite alphabet Σ of size σ**
The graph that contains one vertex for each k-mer (and therefore σ^k vertices) and a directed edge for each overlap of length k − 1 (and therefore σ^{k+1} edges)

Example 2.3





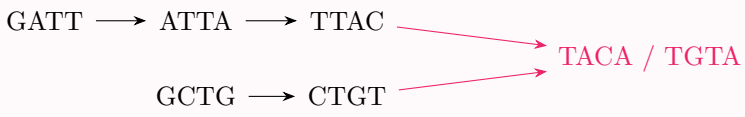
The de Bruijn subgraph for a given sequence s of dimension k is the one that we introduced in Definition 2.1 It has several interesting properties (see Section 2.1) and is used in various contexts in bioinformatics like genome assembly and pangenomics.

Problem 2.1 In most bioinformatics applications where DNA sequence reads come from a sequencing machine, it is not clear from which strand of the DNA double helix a read originates.

$$\text{GATT} \longrightarrow \text{ATTA} \longrightarrow \text{TTAC} \longrightarrow \text{TACA}$$

$$\text{GCTG} \longrightarrow \text{CTGT} \longrightarrow \text{TGTA}$$

Therefore one prefers to identify a k-mer z with its reverse complement $\overleftarrow{z^{-1}}$ and represent them by the same vertex in the graph:



Example 2.4

Definition 2.4 bidirected de Bruijn subgraph

Finally, in a (bidirected) de Bruijn subgraph one often finds stretches of vertices that have only one successor, the next k-mer in the originating sequence(s). Such a non-branching path can be collapsed into a into single vertex (representing a k'-mer for some $k' \geq k$),called unitig, saving space and often computation time.

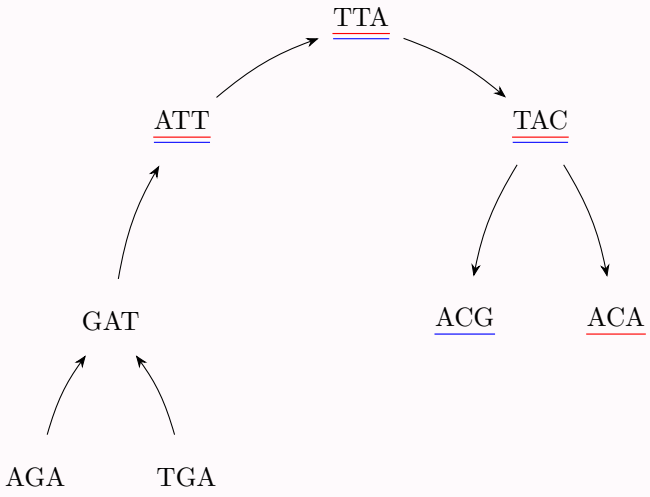
Definition 2.5 compacted de Bruijn subgraph

Example 2.5

An additional variant comes into play when the input sequences are partitioned into several groups, for example representing different genomes, and these groups shall also be distinguished in the de Bruijn graph. Then one assigns a color (usually a single bit indicating presence or absence) to each group. A k-mer inherits this color, so that each k-mer gets assigned a color set, indicating in which groups it appears and in which not.

Definition 2.6 **colored de Bruijn subgraph**

Example 2.6



Problem 2.2 Colored de Bruijn subgraphs form a very powerful data structure, but their memory-efficient implementation is not trivial. In fact, the main challenge is no longer the storage of the k-mers (whose number is limited by σ^k , see above), but the storage of the colors, which may occupy several thousand bits per k-mer in case of large pangenomes.
log der alphabetgröße zur basis 2 ist wie viele bits man braucht zum codieren k*2 bits für dbg, aber mit Farben muss man ja noch bits dran hängen, das ist groß und noch ungelöst, denn es ist dann k*2 + # datensätze

Example 2.7 bsp zu bits: acg codiert als 000110 wenn

A	0	0
C	0	1
G	1	0
T	1	1

2.4 | Implementation of de Bruijn Graphs: Sets of k-mers

Here we consider only the basic variant of de Bruijn subgraphs. With a little bit of effort all results can also be generalized to the other graph types.
A simple observation allows to save considerable space: Assume we have a very fast method to test if a certain k-mer is represented in de Bruijn graph $G = DBG(s, k)$ or not. Then one does not need to store the edges explicitly, for the following reason. When traversing from a vertex representing k-mer au to an adjacent vertex representing k-mer $ub(a, b \in \Sigma, u \in \Sigma^{k1})$, instead of following the edge $au \to ub$ explicitly (if it exists), we can instead just test if ub is represented as a vertex in G or not. If the test is successful, we know that the edge exists and therefore can be traversed. If the test is unsuccessful, then the edge does not exist and can not be traversed. Similarly, if we want to find all possible successors of the vertex representing k-mer au, we perform tests, one for each possible last character c of the new, overlapping k-mer uc. Especially for small alphabets like DNA, this is not very much work.

Therefore, in (DNA-) bioinformatics applications, a data structure is very popular that represents only a set S of k-mers and offers a very quick presence/absence test, instead of storing a complete graph data structure: the Bloom filter.

The idea is to use one large bit array B of length m, say, as a hash table with all bits initially set to $false$ (F), and then have a small number of h different hash functions $f_1, f_2, ..., f_h$ that map k-mers (respectively, their ranks) to integers in the range $[0, m1]$. In order to store a k-mer (respectively its rank r), the hash functions are computed and the corresponding bits in B are set:

$$B[f_1(r)] = B[f_2(r)] = ... = B[f_h(r)] = T.$$

This is repeated for each k-mer.
As always, the hash functions should be independent and uniformly distributed. A very simple one is of the form $f_i(r) = (r^{i+1} >> k) \mod m$, where $>> k$ denotes k-fold bitwise right-shift. There exist, however, much more advanced hash functions in the literature.

In order to test whether a k-mer with rank r has been stored in a Bloom filter B, we perform the same procedure: We evaluate all hash functions in parallel, and if all of them point to T entries in the hash table, then it is very likely that the element r has been stored in B:

$$\text{probably-contains}(B, r) = B[f_1(r)] \wedge B[f_2(r)] \wedge \ \wedge B[f_h(r)],$$

where $\wedge$ denotes logical AND.

By the combination of several distinct hash functions, a k-mer z (with rank r = rank(z)) will be reported as a false positive, although it has not been stored in the Bloom filter, only if all hash functions are involved in collisions, i.e., if for each of the h hash values $f_i(r), 1 \leq i \leq h$, some other k-mer $z' \neq z$ produces with some hash function fi the same hash value $(f'_i(rank(z')) = f_i(r))$ and therefore sets B at this position to T. While this is quite unlikely in practice as long as the hash table is only moderately filled, it can not be completely avoided. Therefore the above function is called “probably-contains” and not “contains”. There are some techniques to handle false positives in certain applications manually, so that they do not distort the result and the Bloom filter becomes exact. Details are omitted here.

Backmatter

Definitionen

q-gram . 5

alignment distance The alignment distance of two sequences $x, y \in \sum^*$ is defined as $d(x, y) := mincost(A) : A \in \mathcal{A} *$ is an alig
5

bidirected de Bruijn subgraph . 12

colored de Bruijn subgraph . 13

compacted de Bruijn subgraph . 12

de Bruijn subgraph . iii, 9–12

dimension of a de Bruijn subgraph . 9

edit distance test. 5

occurrence count Let $x \in \sum^m$ and choose $q \in [1, m]$. The occurrence count of a q-gram $z \in \sum^q$ in x is the number

$$Occ(x, z) := |i \in [1, m - q + 1] : x[i...i + q - 1] = z|$$

. 5

original de Bruijn graph of dimension k over a finite alphabet Σ of size σ The graph that contains one vertex for each k-mer (and therefore σ^k vertices) and a directed edge for each overlap of length k $-$ 1 (and therefore σ^{k+1} edges). 11

q-gram distance The q-gram distance of $x \in \sum^m$ and $y \in \sum^n$ is defined for $1 \leq q \leq min\{m, n\}$ as

$$d_q(x, y) := \sum_{z \in \sum^q} |p_q(x)_z - p_q(y)_z|$$

. 6

q-gram profile Let $\sigma = |\sum|$. The q-gram profile of x is a σ^q-element vector $p_q(x) := (p_q(x)_z)_{z \in \sum^q}$, indexed by all q-grams, defined by $p_q(x)_z := Occ(x, z)$. 6

ranking algorithm . 6

ranking function . 6

unranking algorithm . 6

unranking function . 6

List of Figures